

Cecilia Steel

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in <https://linkedin.com/in/ceciliazsteel/> 🌐 <https://ceciliazsteel.github.io/>

Education

University of California - Davis

- Ph.D in Physics

Davis, CA
Sep 2024 – Present

University of Pittsburgh

- BS in Physics and Astronomy, *with Honors, magna cum laude*
- Minor in Mathematics

Pittsburgh, PA
Aug 2020 – April 2024

Research

Stellar Dynamics in Milky Way–Mass Galaxies

Advisor - Andrew Wetzel

UC Davis
June 2025 – Present

- Investigate stellar dynamics in Milky Way–mass galaxy disks using cosmological zoom-in simulations, with emphasis on quantifying the nature of radial redistribution and the evolution of stellar orbits over time.

Modeling Star Formation Histories of Galaxies at Half the Age of the Universe

Advisor - Rachel Bezanson

University of Pittsburgh
May 2022 – Aug 2024

- Measured empirically motivated star-formation history predictions of massive galaxies from the UniverseMachine model and compared to previously measured star formation histories derived directly from observations.

Survey of Transitioning Extrasolar Planets of the University of Pittsburgh (STEPUP) Member

University of Pittsburgh
May 2022 – Aug 2024

- Member of an undergraduate research team that uses the Keeler Telescope at the Allegheny Observatory. We observe and detect exoplanets, collect and process data on their transits, and generate light curves.

Publications

Hot or Cold? Radial Redistribution of Stars in FIRE Simulations of Milky Way–Mass Galaxies and the Asymmetry of Inward versus Outward Migrators

2026

Cecilia Steel, Andrew Wetzel, Rori Kang, Fiona McCluskey, Sarah R. Loebman, Kathryn J. Daniel
[Steel et al. 2026](#) [🔗](#)

Radial redistribution of stellar orbits in FIRE simulations of Milky-Way-mass galaxies

2026

Matthew A. Bellardini, Fiona McCluskey, Andrew Wetzel, *Cecilia Steel*, Sarah R. Loebman, Jorge Moreno
[Bellardini et al. 2026](#) [🔗](#)

A Comparison of Star-Formation Histories Derived from UniverseMachine and LEGA-C at $0.6 < z < 1$

2024

Cecilia Steel, Alan Pearl, Yashsa Kaushal, Rachel Bezanson
[Steel et al. 2024](#) [🔗](#)

Awards and Honors

Graduate Research Mentorship Fellowship - \$70,000

UC Davis 2026

Ling-Lie Chau Fellowship - \$6,000

UC Davis 2025

Julia Thompson Award For Excellence in Undergraduate Writing	<i>University of Pittsburgh 2024</i>
Sigma Pi Sigma, the National Physics Honor Society Membership	<i>University of Pittsburgh 2024</i>
NASA Pennsylvania Space Grant Consortium Research Scholarship	<i>University of Pittsburgh Summer 2022; Spring, Summer, Fall 2023</i>

Teaching Assistant Appointments

PHY 157 - Astronomy Instrumentation & Data Analysis Laboratory Instructor: Tucker Jones	<i>UC Davis Spring 2025</i>
PHY 7A - General Physics I Instructor: Dina Zhabinskaya & Thomas Weideman	<i>UC Davis Winter 2025</i>
PHY 7B - General Physics II Instructor: Sergey Savrasov	<i>UC Davis Fall 2024</i>

Undergraduate Poster Sessions

Conferences for Undergraduate Women in Physics (CUWIP)	<i>United States Military Academy West Point January 2024</i>
Testing the Star-Formation Histories of Massive Galaxies in the UniverseMachine Model at $z \sim 0.8$ with the LEGA-C Survey ↗	
AAS 243rd Meeting	<i>New Orleans Ernest N. Morial Convention Center January 2024</i>
Testing the Star-Formation Histories of Massive Galaxies in the UniverseMachine Model at $z \sim 0.8$ with the LEGA-C Survey ↗	
University of Pittsburgh Undergraduate Poster Session	<i>University of Pittsburgh November 2023</i>
Testing the Star-Formation Histories of Massive Galaxies in the UniverseMachine Model at $z \sim 0.8$ with the LEGA-C Survey ↗	
Conferences for Undergraduate Women in Physics (CUWIP)	<i>University of Pittsburgh January 2023</i>
(Testing a Theoretical Model (“UniverseMachine”) for Galaxy Star Formation Histories Derived from the LEGA-C Survey) ↗	